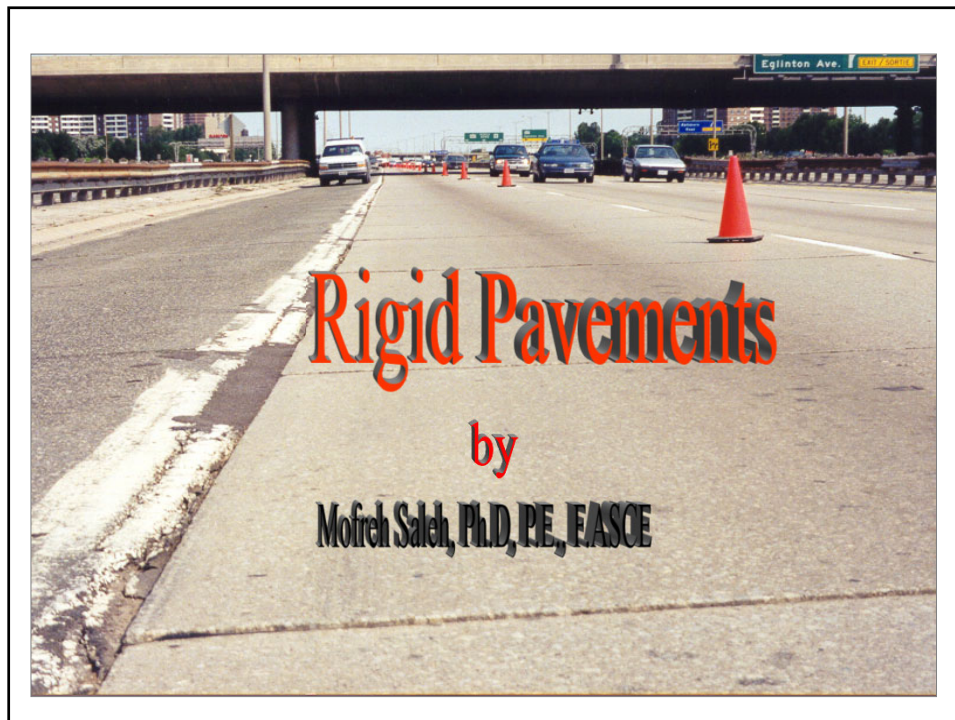




1

Rigid Pavements

Reasons for Stresses in Rigid Pavements:

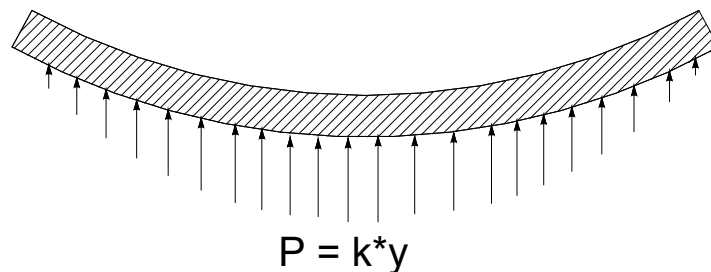
1. Externally applied loads
2. Restrained temperature and moisture displacements
3. Volumetric changes of the supporting material
4. Discontinuity of the subgrade support as affected by permanent deformations and the loss of fines through pumping

2

Modulus of Subgrade Reaction

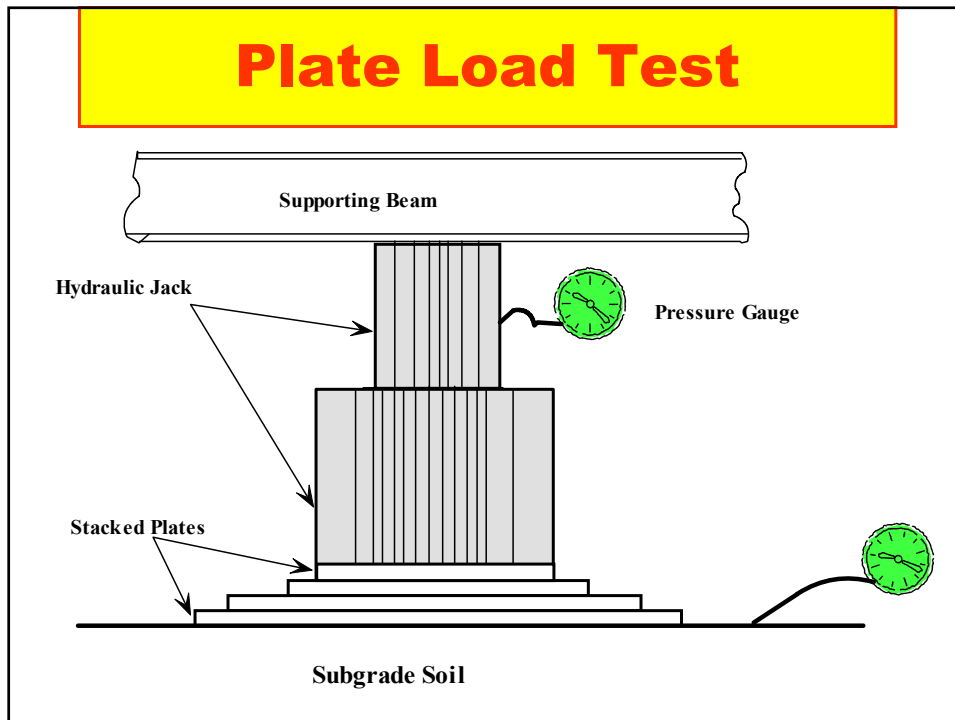
- Assume the case of beam on elastic foundation. It is assumed that the reactive pressure is proportional with the deflection.
- $P \propto y$
- $P = k y$
- P = Applied pressure in kPa
- y = deflection in m
- k = modulus of subgrade reaction $\text{kN/m}^2/\text{m}$ (kN/m^3)
- k is usually determined by plate bearing test.

3

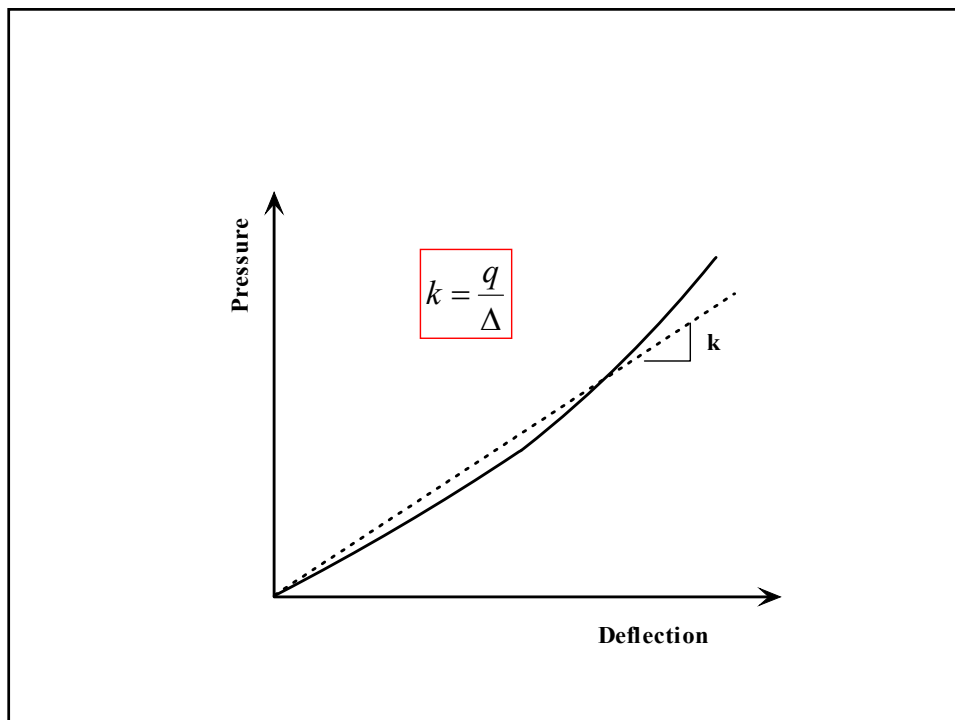


4

Plate Load Test



5



6

Radius of Relative Stiffness

For a beam subject to bending the differential equation relates the moment and deflection is given by:

$$M = EI \frac{d^2 y}{dx^2}$$

The term EI is called the beam stiffness. Similarly, for a slab subjected to moment the differential equation relates the moment and deflection is given by:

$$M_x = \frac{Eh^3}{12(1-\nu^2)} * \frac{d^2 y}{dx^2}$$

The term $\frac{Eh^3}{12(1-\nu^2)}$ is called the slab stiffness.

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The ratio between the slab stiffness and the modulus of subgrade reaction is called the radius of relative stiffness. This term combines the properties of both the slab and the subgrade and has a linear dimension.

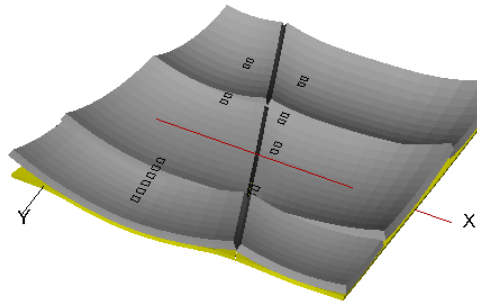
$$\ell = \left(\frac{Eh^3}{12(1-\nu^2) * k} \right)^{\frac{1}{4}}$$

$$\ell = \left(\frac{F / L^2 * L^3}{F / L^3} \right)^{\frac{1}{4}} = L$$

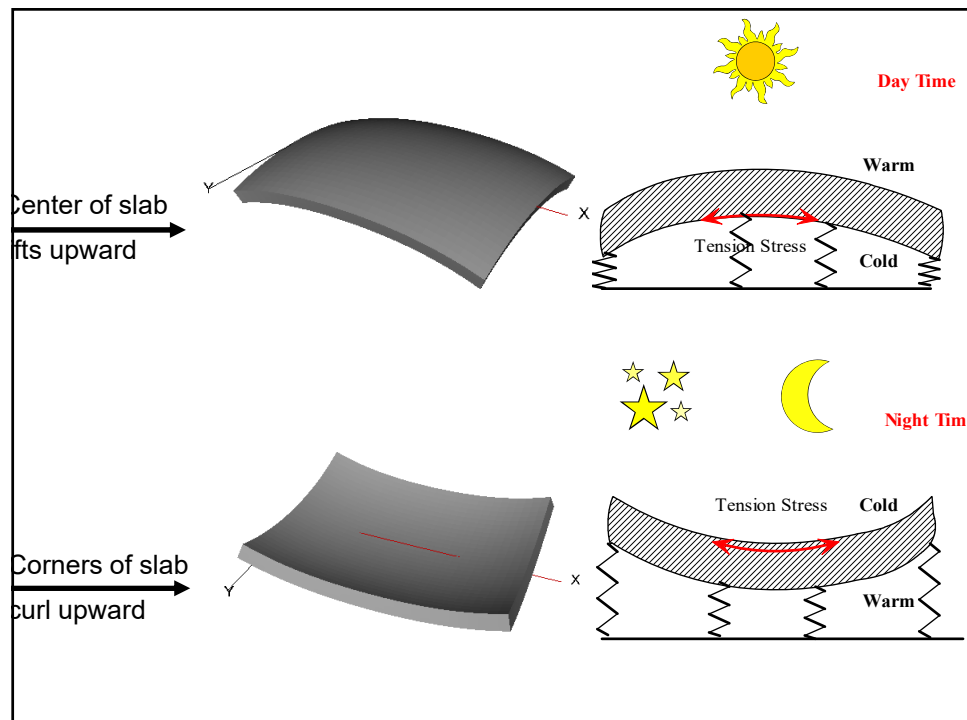
8

Stresses Due to Curling

- If the slab is subjected to temperature differential, it will tend to curl upward or downward and because the weight of the slab will restrain this movement, stresses will develop in the slab.



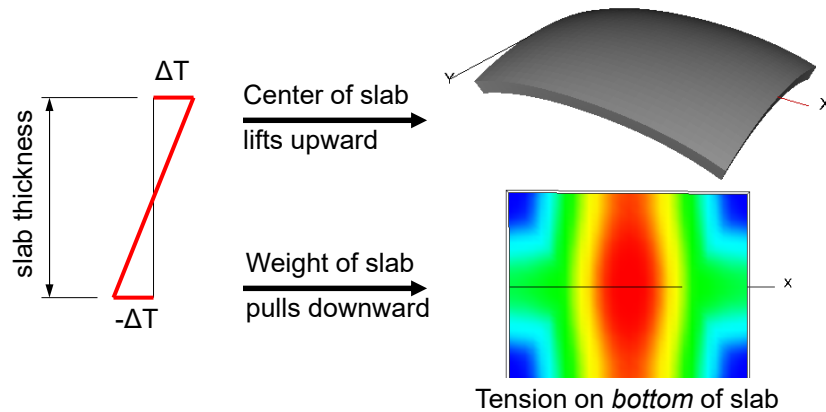
9



10

Analysis of Thermal Gradients

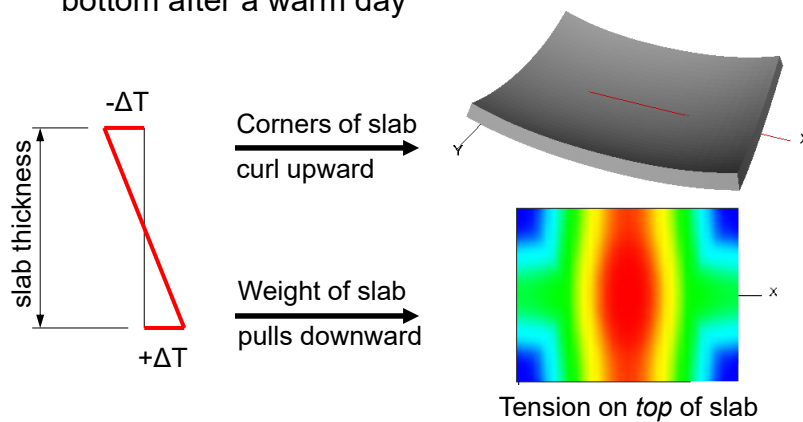
Daytime curling: top of slab heats relative to the bottom during a warm day



11

Analysis of Thermal Gradients

Nighttime curling: top of slab cools relative to the bottom after a warm day



12

Bending of Infinite Slab

$$\varepsilon_x = \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E} \dots\dots\dots(1)$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \nu \frac{\sigma_x}{E} \dots\dots\dots(2)$$

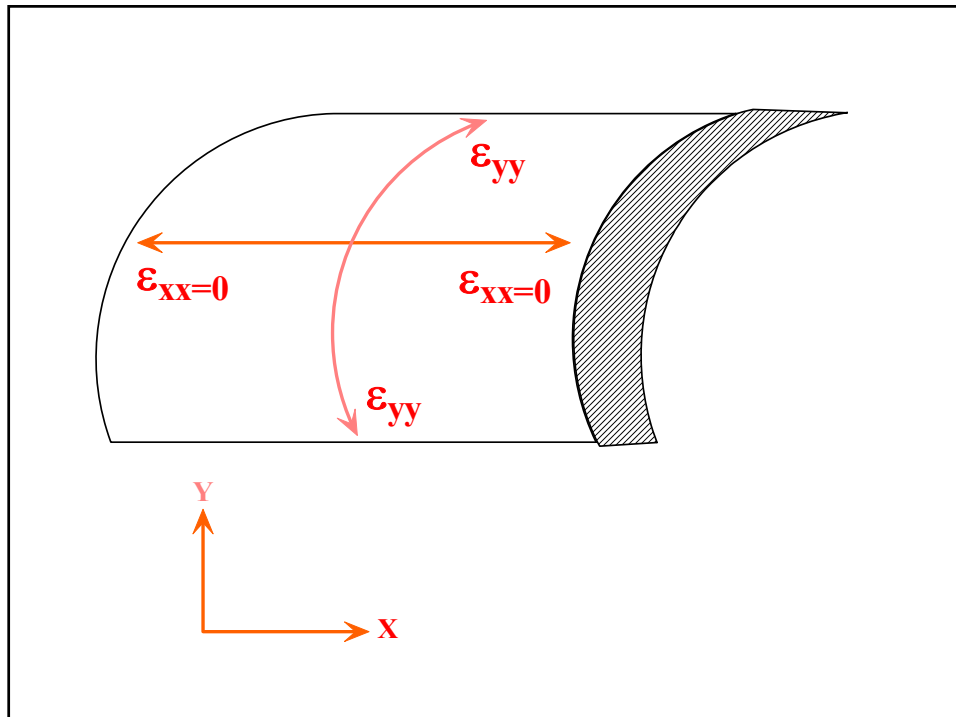
If the slab is restrained so that it will bend only in x direction and because the slab is of infinite dimension $\varepsilon_y=0$, substitute in equation 2

$\sigma_y = \nu \sigma_x \dots\dots\dots(3)$, substitute in equation 1

$$\varepsilon_x = \frac{\sigma_x}{E} (1 - \nu^2)$$

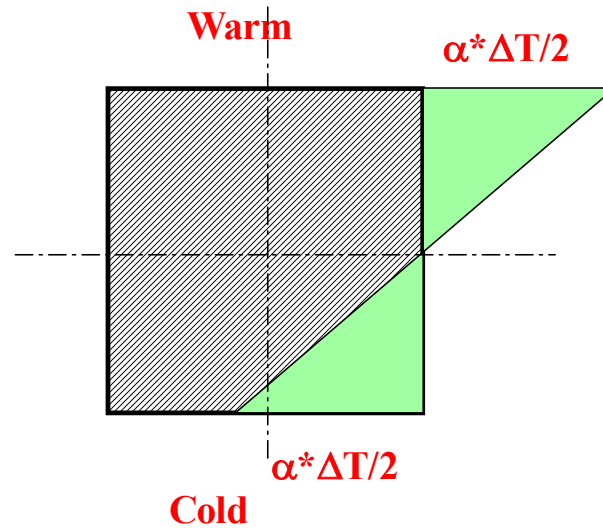
$$\sigma_x = \frac{E \varepsilon_x}{(1 - \nu^2)} \dots\dots\dots(4)$$

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Strains due to temperature



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$$\epsilon_x = \epsilon_y = \frac{\alpha \Delta t}{2} \dots\dots\dots(5)$$

substitute 5 into 4 will yield the stress in x direction due to curling in x direction:

$$\sigma_x = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} \text{ (due to curling in x direction)} \dots\dots\dots(6)$$

$$\sigma_y = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} \text{ (due to curling in y direction)} \dots\dots\dots(7)$$

$$\sigma_x = \nu \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} \text{ (due to curling in y direction)} \dots\dots\dots(8)$$

$$\sigma_y = \nu \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} \text{ (due to curling in x direction)} \dots\dots\dots(9)$$

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Using the concept of superposition, the total stresses in x direction and y direction due to warping in both x and y direction can be determined as follow.

$$\sigma_x = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} + \nu \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} = \frac{E * \frac{\alpha \Delta t}{2} (1 + \nu)}{(1 + \nu) * (1 - \nu)}$$

$$\sigma_x = \frac{E * \alpha \Delta t}{2(1 - \nu)} \dots \dots \dots (10)$$

Similarly,

$$\sigma_y = \frac{E * \alpha \Delta t}{2(1 - \nu)} \dots \dots \dots (11)$$

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Curling Stresses in Finite Slab

- Assume the slab has finite dimensions L_x and L_y

$$\sigma_x = \sigma_x (\text{due to warping in x direction}) * C_x + \sigma_x (\text{due to warping in y direction}) * C_y$$

$$\sigma_x = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} * C_x + \nu \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} * C_y$$

$$\sigma_x = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} * (C_x + \nu * C_y)$$

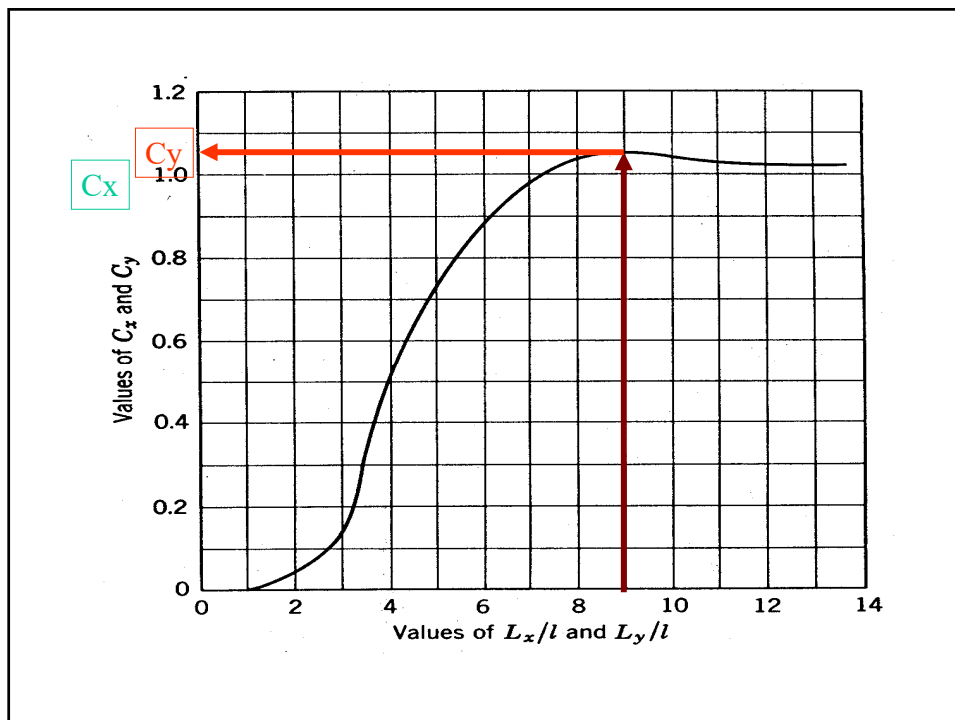
18

$$\sigma_y = \sigma_y(\text{due to warping in y direction}) * C_y + \sigma_y(\text{due to warping in x direction}) * C_x$$

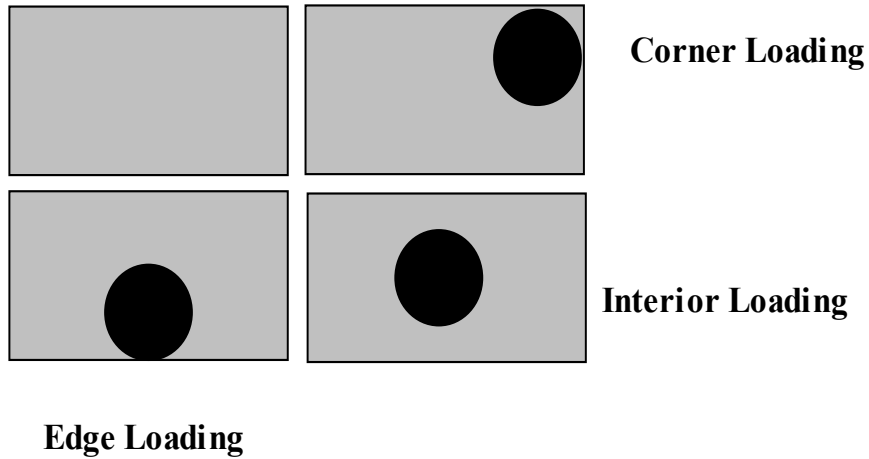
$$\sigma_y = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} * C_y + \nu \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} * C_x$$

$$\sigma_y = \frac{E * \frac{\alpha \Delta t}{2}}{(1 - \nu^2)} * (C_y + \nu * C_x)$$

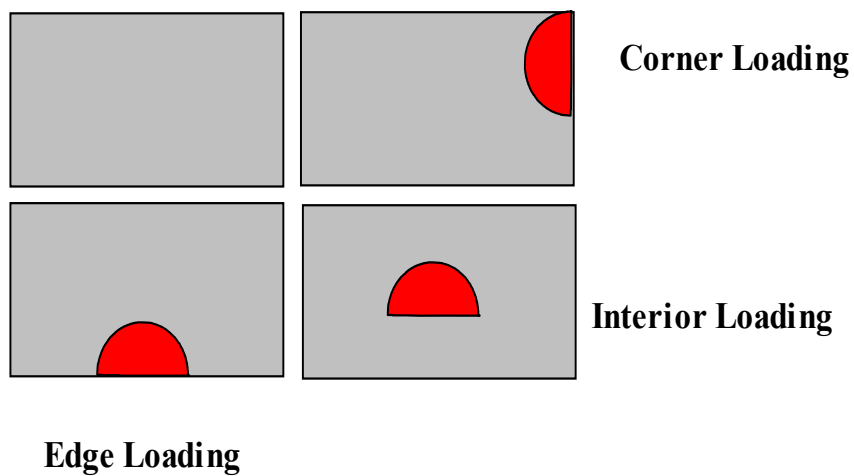
19



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Stresses and Deflection due to Traffic Loading

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Semi-Circular Loaded Area

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Corner Loading**For a concentrated load at the corner**

$$\sigma_c = \frac{M_x}{I} * \frac{h}{2} = \frac{P.X}{(2x) * \frac{h^3}{12}} * \frac{h}{2} = \frac{3P}{h^2}$$

For a circular load at the corner of the slab

$$\sigma_{\text{corner}} = \frac{3P}{h^2} * \left[1 - \left(\frac{a\sqrt{2}}{\ell} \right)^{0.6} \right]$$

$$w_{\text{corner}} = \frac{P}{k\ell^2} * \left[1.1 - 0.88 * \left(\frac{a\sqrt{2}}{\ell} \right) \right]$$

Interior Loading

$$\sigma_{\text{interior}} = \frac{3(1+\nu) * P}{2 * \pi * h^2} * \left(\ln\left(\frac{\ell}{b}\right) + 0.6159 \right)$$

$$b=a \quad \text{if } a \geq 1.724h$$

$$b = \sqrt{1.6a^2 + h^2} - 0.675h \quad \text{if } a < 1.724h$$

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Edge Loading

Two cases will be presented: complete circular load and semicircular load at the edge of the slab.

Circular loaded area at the edge of the slab:

$$\sigma_{\text{edge}} = \frac{3 * (1+\nu) * P}{\pi * (3+\nu) * h^2} * \left[\ln\left(\frac{Eh^3}{100ka^4}\right) + 1.84 - \frac{4\nu}{3} + \frac{1-\nu}{2} + \frac{1.18 * (1+2\nu) * a}{\ell} \right]$$

$$w_{\text{edge}} = \frac{\sqrt{2+1.2\nu} * P}{\sqrt{Eh^3k}} \left[1 - \frac{(0.76+0.4\nu) * a}{\ell} \right]$$

For semicircular Load

$$\sigma_{\text{edge}} = \frac{3 * (1+\nu) * P}{\pi * (3+\nu)h^2} \left[\ln\left(\frac{Eh^3}{100ka^4}\right) + 3.84 - \frac{4\nu}{3} + \frac{(1+2\nu) * a}{2 * \ell} \right]$$

$$w_{\text{edge}} = \frac{\sqrt{2+1.2\nu} * P}{\sqrt{Eh^3k}} * \left[1 - \frac{(0.323+0.17 * \nu) * a}{\ell} \right]$$

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Example

A rigid pavement slab 10m by 3.65 m if the slab has a modulus of elasticity 27.5 GPa, Poisson ratio 0.15, and coefficient of thermal expansion 9×10^{-6} mm/mm/ $^{\circ}$ C subjected to different type of loadings. If the modulus of subgrade reaction is 27200 kN/m³, for each case calculate the following:

Case 1

Temperature differential of 15 $^{\circ}$ C

For this case calculate the curling stresses at the centre of the slab. (Assume the thickness of the slab in this case is 200 mm)

Case 2

Circular load of radius 150 mm and load value of 45 kN is applied at the corner of the slab.

Assume the thickness of the slab is 250 mm, calculate stresses and deflection due to

1. Corner loading
2. Edge loading
3. Interior loading

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Solution**Case 1**

$$\ell = \left(\frac{Eh^3}{12(1-\nu^2) * k} \right)^{\frac{1}{4}} = \left(\frac{27.5 * 10^6 * 0.2^3}{12(1-0.15^2) * 27200} \right)^{\frac{1}{4}} = 0.911 \text{ m}$$

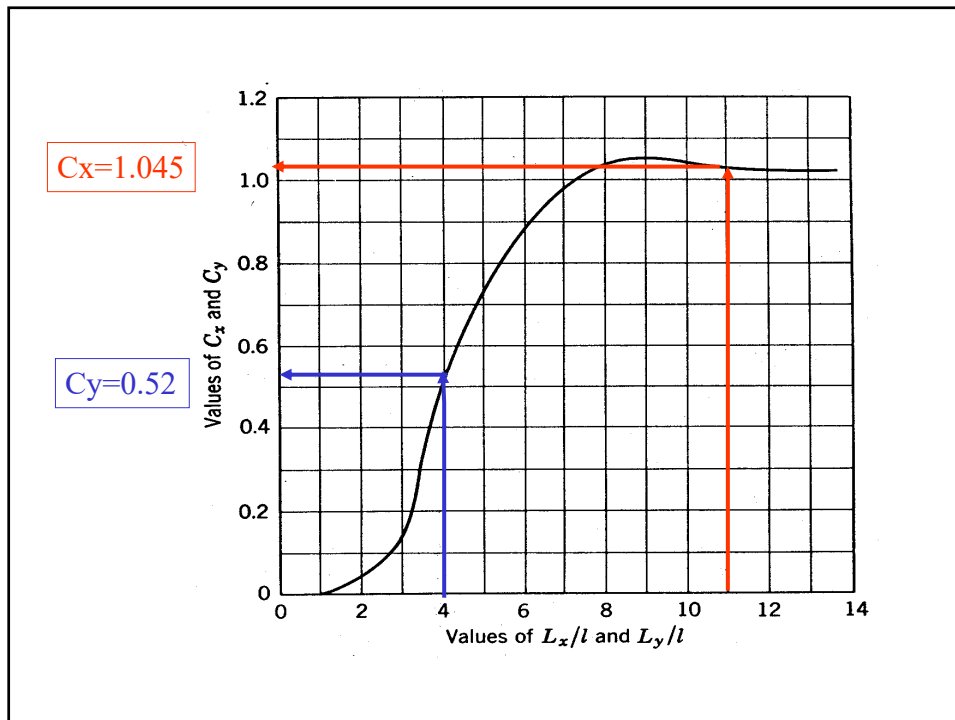
$$\frac{L_x}{\ell} = \frac{10}{0.911} = 10.97 \quad C_x = 1.045$$

$$\frac{L_y}{\ell} = \frac{3.65}{0.911} = 4.01 \quad C_y = 0.52$$

$$\sigma_x = \frac{E * \alpha \Delta t}{(1-\nu^2)} (C_x + \nu * C_y) =$$

$$\frac{27.5 * 10^6 * \frac{9 * 10^{-6} * 15}{2}}{(1-0.15^2)} (1.045 + 0.15 * 0.52) = 2132.55 \text{ kPa} =$$

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Case II

$$\ell = \left(\frac{Eh^3}{12(1-\nu^2) * k} \right)^{\frac{1}{4}} = \left(\frac{27.5 * 10^6 * 0.25^3}{12(1-0.15^2) * 27200} \right)^{\frac{1}{4}} = 1.077 \text{ m}$$

$$a = 0.15 \text{ m}$$

$$P = 45 \text{ kN}$$

Corner Loading

$$\sigma_{\text{corner}} = \frac{3P}{h^2} * \left[1 - \left(\frac{a\sqrt{2}}{\ell} \right)^{0.6} \right] = \frac{3 * 45}{0.25^2} * \left[1 - \left(\frac{0.15\sqrt{2}}{1.077} \right)^{0.6} \right] = 1345.1 \text{ kPa}$$

$$W_{\text{corner}} = \frac{P}{k\ell^2} \left[1.1 - 8.8 * \left(\frac{a\sqrt{2}}{\ell} \right) \right] = \frac{45}{27200 * 1.077^2} \left[1.1 - 8.8 * \left(\frac{0.15\sqrt{2}}{1.077} \right) \right] = -0.903$$

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Case II

$$\ell = \left(\frac{Eh^3}{12(1-\nu^2) * k} \right)^{\frac{1}{4}} = \left(\frac{27.5 * 10^6 * 0.25^3}{12(1-0.15^2) * 27200} \right)^{\frac{1}{4}} = 1.077 \text{ m}$$

$$a = 0.15 \text{ m}$$

$$P = 45 \text{ kN}$$

Corner Loading

$$\sigma_{\text{corner}} = \frac{3P}{h^2} * \left[1 - \left(\frac{a\sqrt{2}}{\ell} \right)^{0.6} \right] = \frac{3 * 45}{0.25^2} * \left[1 - \left(\frac{0.15\sqrt{2}}{1.077} \right)^{0.6} \right] = 1345.1 \text{ kPa}$$

$$W_{\text{corner}} = \frac{P}{k\ell^2} \left[1.1 - 8.8 * \left(\frac{a\sqrt{2}}{\ell} \right) \right] = \frac{45}{27200 * 1.077^2} \left[1.1 - 8.8 * \left(\frac{0.15\sqrt{2}}{1.077} \right) \right] = -0.903$$

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Edge Loading

$$\sigma_{\text{edge}} = \frac{3 * (1+\nu) * P}{\pi * (3+\nu) * h^2} * \left[\ln \left(\frac{Eh^3}{100ka^4} \right) + 1.84 - \frac{4\nu}{3} + \frac{1-\nu}{2} + \frac{1.18 * (1+2\nu) * a}{\ell} \right]$$

$$\sigma_{\text{edge}} = \frac{3 * (1+0.15) * 45}{\pi * (3+0.15) * 0.25^2} * \left[\ln \left(\frac{27.5 * 10^6 * 0.25^3}{100 * 27200 * 0.15^4} \right) + 1.84 - \frac{4 * 0.15}{3} + \frac{1-0.15}{2} + \frac{1.18 * (1+2 * 0.15) * 0.15}{1.077} \right]$$

$$= 2013.55 \text{ kPa}$$

$$w_{\text{edge}} = \frac{\sqrt{2+1.2\nu} * P}{\sqrt{Eh^3k}} \left[1 - \frac{(0.76+0.4\nu)a}{\ell} \right] =$$

$$\frac{\sqrt{2+1.2 * 0.15} * 45}{\sqrt{27.5 * 10^6 * 0.25^3 * 27200}} \left[1 - \frac{(0.76+0.4 * 0.15) * 0.15}{1.077} \right] * 10^3 = 0.544 \text{ mm}$$

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Interior Loading

$$\sigma_{\text{interior}} = \frac{3(1+\nu) * P}{2 * \pi * h^2} * \left(\ln\left(\frac{\ell}{b}\right) + 0.6159 \right)$$

$$b=a \quad \text{if } a \geq 1.724h$$

$$1.724 * h = 1.724 * 0.25 = 0.431 \text{ m}$$

$$a = 0.15 < 1.724h$$

$$b = \sqrt{1.6a^2 + h^2} - 0.675h \quad \text{if } a < 1.724h$$

$$b = \sqrt{1.6a^2 + h^2} - 0.675h = \sqrt{1.6 * 0.15^2 + 0.25^2} - 0.675 * 0.25 = 0.1451$$

$$\sigma_{\text{interior}} = \frac{3(1+\nu) * P}{2 * \pi * h^2} * \left(\ln\left(\frac{\ell}{b}\right) + 0.6159 \right) = \frac{3(1+0.15) * 45}{2 * \pi * 0.25^2} * \left(\ln\left(\frac{1.077}{0.1451}\right) + 0.6159 \right) = 1$$

$$W_i = \frac{P}{8 * k * \ell^2} \left\{ 1 + \frac{1}{2 * \pi} \left[\ln\left(\frac{a}{2\ell}\right) - 0.673 \right] * \left(\frac{a}{\ell}\right)^2 \right\} =$$

$$\frac{45}{8 * 27200 * 1.077^2} \left\{ 1 + \frac{1}{2 * \pi} \left[\ln\left(\frac{0.15}{2 * 1.077}\right) - 0.673 \right] * \left(\frac{0.15}{1.077}\right)^2 \right\} * 10^3 = 0.001621 \text{ mm}$$

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Design of Steel

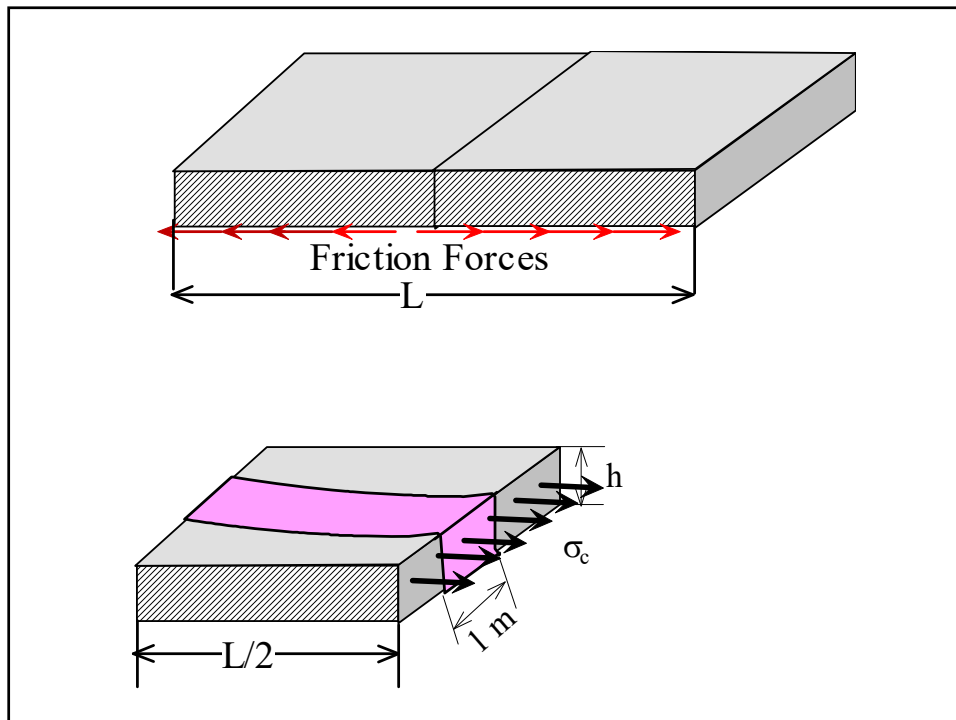
$$\sigma_c * h * 1 = \gamma_c * L / 2 * h * f$$

$$\sigma_c = \frac{\gamma_c * L * f}{2}$$

$$A_s * f_s = \sigma_c * h * 1$$

$$A_s = \frac{\gamma_c * L * h * f}{2 * f_s}$$

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